

GradeCraft: What Can We Learn From a Game-Inspired Learning Management System?

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ABSTRACT

The “gamification” of courses (i.e., designing courses that leverage motivational mechanisms found in videogames) is a movement that is gaining traction in educational research communities and universities. Two game-inspired courses were developed at a high-enrollment public university in an effort to increase student engagement, and to provide students with more personalized learning experiences. We designed a learning management system, GradeCraft, to foreground the affordances of these grading systems, and to enhance the “game-like” experience for students. Along with serving as a translation layer for the grading systems of these courses, GradeCraft is also designed with an eye towards learning analytics, and captures information that can be described as student “process” data. Currently this data includes what types of assignments students choose to complete; how students assign percentage weights to their chosen assignments; how often and how accurately students check or model their course grades; and how successfully assignments are completed by students individually and the class as a whole across a structured grading rubric. We hope GradeCraft will give instructors new insight into student engagement, and provide data-driven ideas about how to tailor courses to student needs.

Categories and Subject Descriptors

K.3.1 [Computers and Education]: Computer Uses in Education – *collaborative learning, computer-assisted instruction, computer-managed instruction*

General Terms

Management, Measurement, Design, Theory

Keywords

Learning analytics, syllabus design, game-inspired instruction, gamification

1. INTRODUCTION

In our experience, discussions around learning analytics often begin with the tongue-in-cheek question: “Where’s the *learning*?” This question points to a real concern within the field, namely, that there can be an overemphasis on the *analytics* (i.e., predictive

models, data mining to detect patterns, etc.) and far too little emphasis on applying this knowledge to contexts where student outcomes can be improved (i.e., the *learning*). Systems like Purdue’s Course Signals represent a step in the right direction because they begin to tackle the open question of “where’s the learning” by providing students with grade feedback that helps them adjust their strategies within a given course to be more successful [2]. We see this work as valuable and worthy of future development, but our approach towards learning analytics is slightly different.

Instead of working within established university courses, we have sought to reify the complex grading systems of non-traditional “game-inspired” courses and have used the opportunity to capture student data that speaks to the process through which they reach course outcomes. In so doing we deliberately sit at the nexus of learning analytics and “gamification.” While these data are generally descriptive in nature at this point, we see this as a subject for iterative design; as courses evolve, so too will our understanding of the data that we need to collect and analyze.

Why “game-inspired,” however? “Good” games typically inspire players to spend large amounts of time and effort achieving in-game success. Well-designed games succeed because they tap into our deep-seated desire to learn and be engaged. They give players multiple ways to succeed, maximize choice, and mitigate the cost of failure [4]. Educators over the last decade have been inspired by the depth of content learned and the high-intensity effort that gamers choose to put in when engaged in a good game [6]. Similarities that commonly exist between games and school include well-defined goals at the outset, the establishment of specific challenges to be conquered, requiring practice to succeed, and using assessments to gauge whether material has been properly learned. These parallels led to the question of whether school itself could be made into a good game.

We are currently seeing the first wave of game-inspired courses at our university. These courses aim to increase student choice while mitigating the negative impact of failure. Gameful elements we have observed include: using points and incremental levels instead of grades; awarding badges to recognize achievements and skill-acquisition; allowing students to redo certain assignments as many times as necessary to succeed; and giving students the ability to decide what types of assignments they will take on and how much those assignments will be worth. Thus far our research indicates that the frame shift of these grading systems is motivating and encourages students to complete more work within the course [3].

Given the complexities and choices inherent in a game-inspired grading system, it can be difficult for students to quickly and intuitively grasp the options available to them, and also tricky for instructors to manage. To alleviate these issues we developed

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LAK '13, April 08 - 12 2013, Leuven, Belgium

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GradeCraft, a learning management system designed for the purpose of foregrounding the affordances of gameful grading systems. This design brief describes the current application and the nascent Learning Analytics potential of GradeCraft.

1.1 Case Study: Videogames & Learning

GradeCraft was initially developed to support an elective class on Videogames & Learning at a high-enrollment American public university. The goal of this course is to examine the learning and motivational theories that operate within—and inform the design of—videogames. Rather than treat this content abstractly, the course grading system is infused with gameful design principles. The course has required assignments in addition to a series of optional assignments. While completing *all* of the assignments is possible, it involves more work than is necessary to do well in the course. The variety of assignment options is an example of giving students multiple routes to success—a key principle that makes videogames motivating [4].

These assignments are divided into three categories: “Grinding” assignments are characterized as necessary for learning the content, but are not always as engaging as other assignments, including attendance, weekly reading reactions, blogging, and team activities; “Learning From Playing A Game” assignments center on students reflecting and commenting on a commercial videogame they have chosen to play throughout the term—their “game text”—and are required for all students; “Boss Battle” assignments are longer, more complex, and require a certain level of content mastery to complete successfully. As a result, Boss Battle assignments occur near the end of the term.

Optional assignments in this course may be used as a pathway to exceed the course’s main requirements (because a student wants an “A+,” perhaps) or as a way to regain points that were lost as a result of being absent from class, missing a reading reaction, or simply performing poorly on a required assignment. This latter path is an example of how a game-inspired grading system can mitigate the cost of localized “failure” in coursework, thereby allowing students the opportunity to self-correct, complete the stated learning goals, and succeed in the overall course.

Badges were introduced in the third iteration of the course to highlight specific skills the instructor wanted students to learn, but earning them did not contribute value towards the final grade.

1.2 Case Study: Intro. to Political Theory

GradeCraft is also currently being used in a political science course at the same university. The professor, hoping to increase student motivation and encourage mastery-driven learning, designed the following gameful grading system:

Forty percent of a student’s final grade is “traditional” in that it consists of a core set of requirements: attending lectures, writing weekly reading reactions, and attending teaching assistant led discussion sections. These requirements are designed to provide a core set of content knowledge and a common course experience.

The remaining sixty percent of a student’s grade is determined by two student-driven decisions:

First, students must choose what types of assignments will make up the remaining sixty percent of their grade. There are four assignment types to choose from during the term: traditional essays, an open-ended group project, an open-ended individual project, and contributing to the class blog. Students are encouraged to work on two of the four assignment types, but are allowed to select any number.

Second, students are given the freedom to determine how each of the four assignment types is weighted. This decision is operationalized by giving students six points to “spend” on any assignment type they wish. These points determine the weight of each assignment they select and have no inherent value—they are only meant to simplify the process of determining how the assignments they choose to do are weighted. A student can, for example, assign all six points to academic essays. Doing so amounts to the student choosing to focus only on writing essays.

Finally, students are awarded badges that either recognize or incentivize certain behaviors. These badges are valued at up to twenty-five percent of the student’s grade.

The remainder of this design brief will explore how these choice-driven grading systems are reified through the GradeCraft’s student and instructor interfaces as well as explore the implications for the kind of data collected by GradeCraft.

2. GRADECRAFT

GradeCraft allows for three types of users: students, teaching assistants, and instructors. For the purposes of this paper we will collapse the teaching assistant and instructor roles, as they are functionally similar. At its core GradeCraft is a comprehensive dashboard that allows students to see their course performance in a single view, much like the dashboard of a videogame. The instructor dashboard displays a summary of students, sections, and the overall class performance across a variety of metrics.

We will illustrate the analytics displayed through a series of hypothetical situations, beginning with reviewing what a student sees in GradeCraft.

2.1 The Student Perspective

Upon logging into GradeCraft, a student sees their current score, a chart of the points they have earned so far, and a chart of the points that are available to earn throughout the entire course.

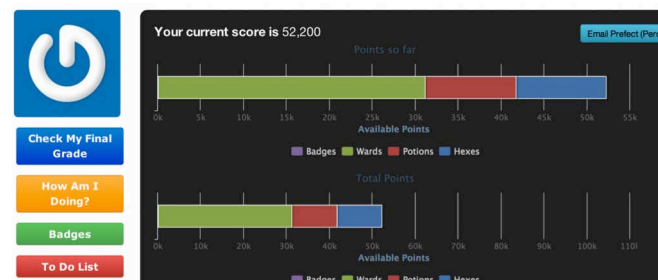


Figure 1: Student Dashboard

The dashboard shows students up-to-date grade information for their course. These descriptions are intended to be both informative and empowering because students have a summary view of their performance, and see a decomposed view of performance by assignment type (Figure 1). This information can be buried in traditional courses, and its inclusion reflects a growth model of learning—students are shown that their current progress can be altered through effort and forward planning for future assignments.

On their dashboard, students can also see which badges they have earned, their progress towards completing unearned badges, and which badges their classmates have earned. They can notify instructors that they have completed a specific task on the path to earning a badge.



Figure 2: Badges for “understanding” course content

Course objectives are also displayed on a student's dashboard, allowing them to visualize their progress towards these overarching goals. This helps orient students to the greater purpose of the course, rather than just their grades.

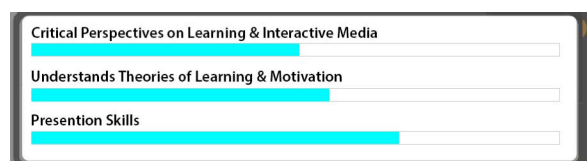


Figure 3: Visualizing Learning Objectives

To that end, one goal we have is for GradeCraft to encourage students to be less concerned with grades and more focused on mastering skills by completing assignments and earning badges. To this end, we have chosen not to display course grades on the student dashboard by default. However, we recognize that we are working within a dominant educational paradigm where grade information is highly sought after, so students can click the "Check My Final Grade" button in order to see how their current score would translate into a final course grade. This information is recorded, and allows instructors to know which students are focused on this outcome. Such information could inform future course design or pedagogical strategies.

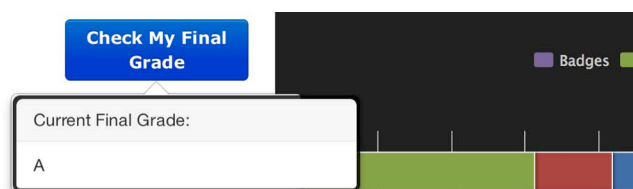


Figure 4: Checking Their Grade

Students can also check how their current score compares to the class average by clicking "How Am I Doing?" (Figure 5). This displays a box-and-whisker plot intended to help students gauge their performance against the rest of the class. We opted not to use “leaderboards” because they can be de-motivating to low-ranked students/players.

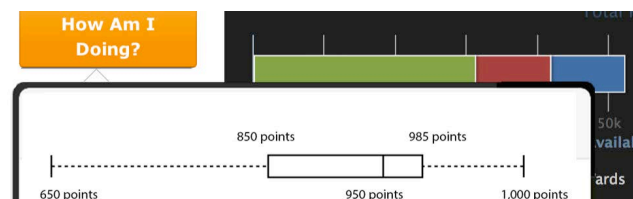


Figure 5: Checking Class Standing

We believe giving students meaningful choices (ones that have an impact on how they will “play the game”) is crucial to designing a good game [5][7]. Students in the political science course mentioned above are able to select which assignments they will do and how much of their grade those assignments will be worth. But how do they make these complex calculations? We built a grade prediction tool to help them in this process. When the predictor is first loaded, the bar chart fills with any points they've already earned, broken down by assignment type.

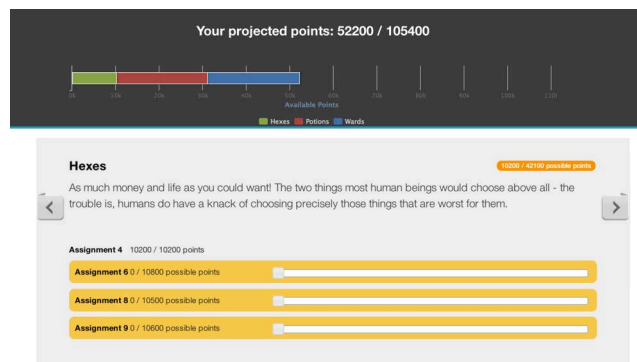


Figure 6: The Grade Predictor

The student can page through each assignment type, selecting how many assignments they plan to do, how well they believe they will score on them, and if applicable, how they would like to weight that assignment type. Students are thus able to model their performance, taking more responsibility for their choices than is typical in college syllabi. GradeCraft captures these modeling instances, allowing instructors and researchers to begin to understand the process behind how students make course engagement decisions.

The predictor stays in sync with the student throughout the semester, showing them their current score at any given moment, and allowing them to assess what work must be done to earn a particular grade. The predictor also acts as a resource, displaying links to materials and tools recommended by the instructor to help students complete the assignment.

2.2 The Instructor Perspective

When initially setting up a course in GradeCraft, instructors can declare the overarching course objectives. As they create assignment types and badges, each one can be tagged with the relevant course objectives.

Instructors can then visualize exactly how their objectives are distributed across the entire course per activity, and in relation to the grading scheme. This helps instructors to see if their learning objectives are fully represented in the course structure or if there are elements that need clearer application.

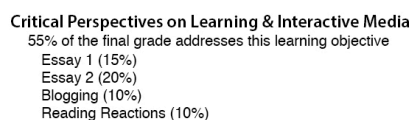


Figure 7: Learning Objectives Assignment Breakdown

Instructors can create badges to encourage the development of particular attributes, skills, or actions that they feel are important for their students to have or do. Each badge has a set of criteria that must be accomplished in order for a student to earn it. Instructors can easily view which badges have been earned, how often, and when. They can also see which badges students are working on, which criteria have been marked complete, and which are proving more difficult for students to achieve.

	Blogger	Learning from Mistakes	Scholar	Discussion Leader	Level 1
Player					
Rita Bennet			50% complete!		
Kimberly Blake	20% complete!				

Figure 8: Badge Analytics

Standard access data—login count, page views, resources accessed—provide the basic framework within which instructors can first begin to investigate student engagement. Instructors can view an interactive table displaying student and class statistics, on these metrics. While this data is possible to collect in all learning management systems, displaying these metrics for instructors' use is not frequently taken advantage of. Making this data plainly available to instructors allows them to have a richer understanding of how their students are choosing to engage—or disengage, as the case may be—with their course. With this information instructors can intervene as necessary to improve student outcomes.

Name	Score	Badges Earned	Log Ins	Page Views	Predictor Views	Last Self-Predicted Grade	Current Final Grade
Rita Bennet	52200	18	21	38	21	A+++	A
Kimberly Blake	50070	12	10	15	3	A	A
Irene Hoffman	45870	8	10	18	4	A	B
Scott Ricker	44919	7	14	21	5	B+	B

Figure 9: Student Activity

When approaching the mid-point of the semester, a professor viewing the engagement charts could, for example, sort students by their attendance record and schedule conversations with students who have missed a high percentage of classes. Alternatively, an instructor might notice that a student who had been performing well in the course suddenly stopped attending and has not been turning in assignments. This acts as an early warning system, suggesting to the instructor that something has changed and that they should contact the student while it is still relatively easy for the student to recover from the situation.

Students also need feedback regarding their class performance and guidance as to what else they *should* be doing. GradeCraft gives students direct access to analytics that can help answer these questions, and also provides instructors with further material to support conversations with students regarding what additional work they can do in the course. Instructors can see each student's dashboard view, visualize how well they have completed the comprehensive learning objectives, and check where the student has ranked in completing each assignment.

In discussing with students what their specific path through the course is, instructors can use the predictor tool to keep track of the choices a student has made, and make recommendations regarding future work. GradeCraft logs the choices students make in the predictor and what final grade these selections ultimately produce. Instructors can see students' predicted final grade charted over time, and drill down to explore which specific assignments a student was considering doing, and how well they expected to do on any particular assignment.

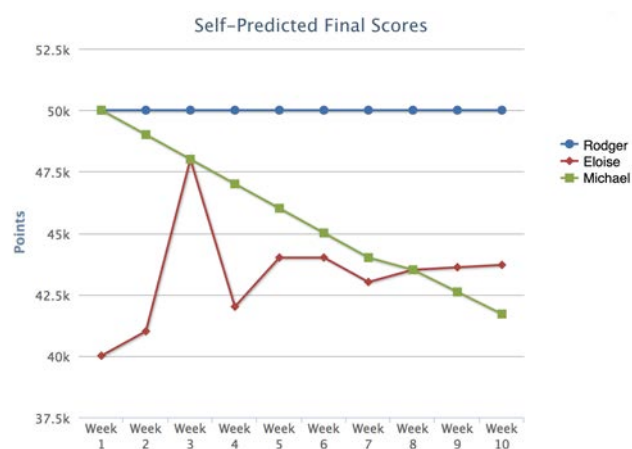


Figure 10: Self-Predicted Final Grades Over Time

If instructors know when students deviate negatively (through missing assignments or lower than expected scores) from their original intentions on an assignment, they can predict earlier in the course timeline which students may be in need of support. This is an improvement on current early warning systems, which rely on comparing a student's behavior to previous classes or current classmates. The unit of analysis in GradeCraft is the course itself, rather than the academic histories students bring with them. These histories cannot fully capture the nuance of a student performing differently than they themselves had *intended*.

GradeCraft allows instructors to visualize which assignment types students choose to complete, and how much weight they decide to assign them. Mapping these choices back to students' final grades will help us investigate if students know their own skills and choose to weight them higher, or if students weight things so as to reduce the risk of working on assignment types they are less familiar with. Given that one of the long-term goals of game-inspired grading systems is to encourage students to explore new types of activities and broaden their skill sets, tracking this relationship will be key to understanding if the frame change is successful in achieving this goal. We need to understand how students perceive risk-taking in relation to assignment type selection, and how to incentivize this behavior to achieve the best learning outcomes.

Clearly-designed rubrics have the advantage of helping students understand explicitly what is expected of them and how they should direct their efforts, provide a more concrete avenue for instructor feedback, and reduce bias in grading [1]. To support these goals, we designed an interactive grading rubric, allowing instructors to set categories, enter criteria and scoring guidelines, and then select how students had fared on each criterion. These assessments build to a final score on the assignment.

From these selections instructors can then explore how well students complete their work from the perspective of these criteria. An instructor might create the category of “Writing Skill” in a long-form essay rubric, with the criteria of “Spelling & Grammar,” “Clarity of Thought,” and “Supported by Research.” When the grading has been completed, the instructor can then see how students individually, as a section, and as a class did on each criterion. Different levels of success would have implications for student, section, and class interventions, explanations, and greater understanding of what the class as a whole needs to be taught in order to succeed.

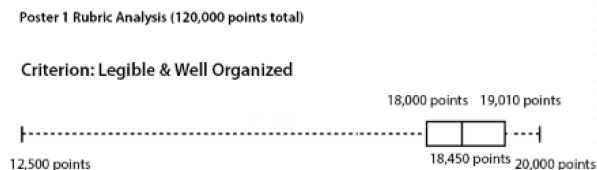


Figure 11: Class Criterion Score for Poster Assignment

3. FUTURE WORK

To date the development of GradeCraft has been focused on the construction of the core LMS functionality and the gathering of data to power basic learning analytics. Our next task is to take the analytic information on student behavior and return it to the instructor and the student in more meaningful, nuanced ways. Currently an instructor can load the profile of any student to check their raw activity statistics. However, especially in large courses, this effort is both unwieldy to perform and a simplistic analysis at best. Therefore, we intend to build a tool that allows the instructor to establish a series of metrics that singly, or in combination, create grounds for identifying students “of concern,” i.e., those with specific characteristics that suggest the desirability of review and potential contact by the instructor.

In keeping with the gameful nature of the system, we also intend to build a “health bar” for students to review course progress. This tool could be set to mirror the instructor’s early warning system metrics, or be based on entirely separate criteria. For instance, an instructor might choose to have the students’ health bar be fifty percent reflective of attendance and fifty percent based on their performance across course learning objectives. Absence from class would be marked by diminished health, which could be regained by doing other optional activities like blogging. Poor performance on learning objectives, as assessed via the interactive grading rubrics, would also decrease a student’s health. Revising

and resubmitting the assignment in question, or completing an alternative assignment to show improvement on that learning objective, would serve to boost the student’s health bar.

4. CONCLUSIONS

GradeCraft gives educators access to new types of analytics that are focused on *learning* because of the multiple points at which students interact with the system. Rather than simply “drilling down” into a letter grade and examining the parts that constitute it, GradeCraft collects data that speaks to process and decisions. It captures which assignments students choose to complete, how students weight those assignments, how students did in regards to completing specific assignment rubrics, and which badges were awarded throughout the course. Each of these decisions is captured, and it is our hope that the resulting data will yield valuable insights about student behavior within game-inspired courses. It is our hope that GradeCraft foregrounds the affordances of the game-inspired grading systems in such a way as to make complicated decisions clear, while also yielding data that speaks to student processes, as opposed to simply reporting student outcomes.

5. ACKNOWLEDGMENTS

We would like to acknowledge Mika Lavaque-Manty for his contributions to the design of GradeCraft, as well as the USELab at the University of Michigan for their feedback and support.

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